# THE CASE FOR LTO AT PETABYTE SCALE

Single-copy cold archive vs online RAID6 | Revised 21 October 2026

**Recommendation.** Use LTO as the retention tier for large, mostly immutable VMS archives when retrieval is infrequent and delayed access is acceptable. Keep the active recording workload and recent footage in a separate online tier. The strongest case is simpler capacity growth, substantially less powered hardware, and lower ongoing media-maintenance exposure.

This report evaluates **1, 5, 10, 20 and 50 PB**, a **single tape copy**, and **two or three compatible LTO-9 drives**. The drives provide access redundancy; they do not duplicate the stored video or recover an irrecoverably damaged sole cartridge.

| At 10 PB retained | Online HDD RAID6 | Single-copy cold LTO-9 |
|---|---|---|
| Storage media | 580 × 24 TB HDDs | 618 × 18 TB cartridges |
| Powered data-access devices | 580 HDDs | 2 or 3 tape drives |
| Device failures/year, rated MTBF | 2.03 HDD failures | 0.070 / 0.105 drive failures |
| Drive-only annual energy | 32.0–39.6 MWh reference range | 0.30–0.87 MWh reference range |
| Growth of stored capacity | More HDDs / groups / bays | Add cartridges; old data need not move |
| Protection from media failure | Two parity members per group | No second video copy |

Energy figures compare the specified devices only: they omit enclosures, robotics, servers, cooling, UPS losses and archive-room conditioning. Failure counts exclude tape media and all infrastructure. Thus these rows demonstrate operating burden, not an equivalent data-loss probability. [1-3]

## What the evidence supports

At 10 PB, the specified component MTBFs imply roughly **19–29 times fewer data-access device failure events** with 3 or 2 tape drives than with 580 HDDs. The media stored on shelves remain passive. This is a defensible advantage; claiming that a single tape copy is necessarily more durable than RAID6 is not.

## The condition that accompanies the recommendation

A failed HDD is normally reconstructed from parity. A failed sole cartridge may permanently lose its contents. Public LTO specifications do not provide a validated cold-storage annual cartridge failure rate. The report therefore gives explicit sensitivity calculations and required-rate thresholds, not a fabricated tape AFR.

The numerical model concerns committed, initially verified archive data. It excludes initial recording loss, ransomware, disasters, catalog/key loss and availability failures outside the drive set. Those exclusions must be addressed in implementation. All capacities use decimal PB/TB.

# 1 / Capacity and hardware-failure model

| Parameter | Baseline |
|---|---|
| C | Retained logical video capacity, TB; 1 PB = 1,000 TB |
| $d_H$, n, $f_H$ | 24 TB HDD; 10 members per RAID6 group; 90% allocatable after parity |
| $d_T$, $f_T$ | 18 TB native LTO-9; 90% planned allocation; one video copy |
| HDD model / MTBF | Seagate IronWolf Pro ST24000NT002 / 2,500,000 powered hours |
| D / tape-drive MTBF | 2 or 3 HPE LTO-9-class drives / 250,000 powered hours |
| Operating state | All HDDs online; tape cartridges stored cold; tape drives assumed powered 24/7 |

$$g(C)=\left\lceil \frac{C}{f_H(n-2)d_H} \right\rceil, \quad N_H(C)=ng(C)$$

$$N_T(C)=\left\lceil \frac{C}{f_T d_T} \right\rceil, \qquad D\in\{2,3\}$$

$$\lambda_H=\frac{8760}{2{,}500{,}000}=0.003504, \quad \lambda_D=\frac{8760}{250{,}000}=0.03504$$

$$\Lambda_H(C)=N_H(C)\lambda_H, \quad \Lambda_D=D\lambda_D$$

$$\mathbb{E}[F(T)]=\Lambda T, \qquad P(F(T)\geq 1)=1-e^{-\Lambda T}$$

Rates are failures per component-year; Λ is failures per system-year. Constant hazard and prompt replacement maintain the installed fleet. The probability formula additionally assumes independent Poisson failures. MTBF is a population rating, not a guaranteed component lifetime.

| Retained | HDDs | Cartridges | HDD failures /yr | HDD P(≥1) /yr |
|---|---|---|---|---|
| 1 PB | 60 | 62 | 0.210 | 18.96% |
| 5 PB | 290 | 309 | 1.016 | 63.8% |
| 10 PB | 580 | 618 | 2.032 | 86.9% |
| 20 PB | 1160 | 1235 | 4.065 | 98.28% |
| 50 PB | 2900 | 3087 | 10.162 | 99.996137% |

**Tape-drive results at every capacity:** two drives yield 0.0701 failures/year and a 6.77% annual chance of at least one drive failure; three yield 0.1051 and 9.98%. More drives increase maintenance events but improve service resilience. Cartridge failures are not included.

The HDD rating depends on vendor conditions and a five-year service-life basis. At an assumed 2% HDD failure intensity, maintenance becomes 1.2, 5.8, 11.6, 23.2 and 58 replacements/year across the five sizes. This is a stress scenario, not a measured rate for the selected model. [1,2]

# 2 / Retention loss: single-copy tape

Let q be the cold-storage annual hazard of a complete cartridge becoming irrecoverably unreadable after successful writing and verification. q is **unknown** for the purchased population and conditions. Neither the tape-drive MTBF nor a 30-year archival-life specification determines it. [3,4]

$$p_T(T) = 1 - e^{-qT}$$

$$P_{\text{loss, tape}}(C, T) = 1 - e^{-N_T(C)qT}$$

$$\mathbb{E}[F_T(T)] = N_T(C)(1 - e^{-qT}), \quad \mathbb{E}[C_{\text{lost}}(T)] = C(1 - e^{-qT})$$

T is in years. These formulas assume independent, constant cartridge hazards and no replacement before retrieval. The final formula assumes whole-cartridge loss and the same hazard for every stored byte. They exclude localized unreadable files, which require an additional model.

| Retained | q = 0.001% /yr | q = 0.01% /yr | q = 0.1% /yr |
|---|---|---|---|
| 1 PB | 0.06198% | 0.6181% | 6.012% |
| 5 PB | 0.3085% | 3.043% | 26.58% |
| 10 PB | 0.6161% | 5.993% | 46.1% |
| 20 PB | 1.227% | 11.62% | 70.92% |
| 50 PB | 3.04% | 26.56% | 95.44% |

**Table: probability of at least one lost cartridge in one year. All three q values are hypothetical sensitivity inputs, not claims about LTO.** A larger archive has more opportunities for an isolated loss even if the lost fraction stays small.

Example: at 10 PB and assumed q = 0.01%/year, the year-one probability of a lost cartridge is 5.99%, while expected lost volume is about 1.00 TB/year. This is an ensemble average: an individual event can lose roughly a cartridge of footage, not a predictable 1 TB each year. Two or three tape drives do not reduce this term.

## Specify the retention objective before claiming sufficient reliability

$$q_{\max}(C) = \frac{-\ln(1-\varepsilon)}{N_T(C)} \quad \text{for a one-year target } \varepsilon$$

For an illustrative 1% annual archive-loss objective, q must be no more than **0.01621%** per cartridge-year at 1 PB, **0.001626%** at 10 PB, or **0.0003256%** at 50 PB.

The 1% objective is an example, not a recommended retention SLA. A no-loss requirement for evidentiary footage may be incompatible with single-copy retention. Storage conditions, media qualification and verification records must support the selected q or upper bound.

# 3 / RAID6 durability and read-error context

Use 10-drive groups with two parity members and 72-hour mean restoration time. For independent whole-drive failures, one active repair per group, exponential repair time and rare failure events:

$$r_{H,3}(C) \approx g(C) n(n-1)(n-2)\lambda_H^3 \tau^2$$

$$\tau = \frac{72}{8760} \text{ years}, \qquad P_{H,3}(T) \approx 1 - e^{-r_{H,3}(C)T}$$

| Retained | Triple-failure loss rate /yr | At assumed HDD λ = 2% /yr |
|---|---|---|
| 1 PB | 1.256e-08 | 2.335e-06 |
| 5 PB | 6.068e-08 | 1.128e-05 |
| 10 PB | 1.214e-07 | 2.257e-05 |
| 20 PB | 2.427e-07 | 4.514e-05 |
| 50 PB | 6.068e-07 | 1.128e-04 |

These tiny numbers are **one mechanism only**. They omit latent sectors, error clustering, shared faults and operational mistakes. They cannot establish total RAID6 durability. Empirical disk research has found correlations and departures from simple independent exponential models. [5]

## LTO has a strong specified read-error advantage

The selected HDD manual specifies at most one nonrecoverable sector per $10^{15}$ bits read. HPE quotes $10^{-19}$ uncorrected error rate for its tape drive family. The LTO-9 format analysis discusses better than $10^{-20}$. Use the contracted product rating, not the most favorable general claim. [1,3,6]

$$\mathbb{E}[\text{read errors}] = 8 \times 10^{12} R u$$

$$P(\text{read errors} \geq 1) \approx 1 - e^{-8 \times 10^{12} R u} \qquad (R \text{ in TB})$$

| One complete read | HPE LTO u = $10^{-19}$ | LTO format u = $10^{-20}$ |
|---|---|---|
| 1 PB | 0.07997% | 0.008% |
| 10 PB | 0.7968% | 0.07997% |
| 50 PB | 3.921% | 0.3992% |

Table values are Poisson calculations at specified rates, not measured probabilities and not annual retention AFR. RAID parity can repair HDD read errors; single-copy tape has no second copy for an irrecoverable error. Thus the 10,000–100,000× component read-error specification difference is **not** a proven system-durability multiplier.

For this report, prefer the supported argument: tape avoids continuously powered media mechanics and has strong built-in error correction; RAID6 provides repairable media redundancy. Their overall durability cannot be ranked numerically from the public specifications alone.

# 4 / Two or three tape drives: what they solve

**Two compatible drives:** one can continue retrieval or ingest after the other fails, provided the surviving drive meets demand. **Three:** two remain after one failure, leaving more capacity for parallel operations and maintenance. Alternate-drive read attempts can resolve some drive-specific problems, but cannot reconstruct genuinely lost media.

| Property | Two drives | Three drives |
|---|---|---|
| Expected drive failures/year | 0.0701 | 0.1051 |
| P(at least one fails/year) | 6.77% | 9.98% |
| Normal planned streaming capacity | 420 MB/s | 630 MB/s |
| Capacity after one drive failure | 210 MB/s | 420 MB/s |
| Post-failure transfer budget/year | 6.62 PB | 13.25 PB |

Throughput assumes 300 MB/s native per drive and a 70% planning factor. The budget combines ingest, verification, migration and retrieval. It is not a promise for small random requests. Compatible media/software, failover routing and prompt repair are required. [3]

$$b_{\text{ingest}} + b_{\text{read}} + b_{\text{verify}} + b_{\text{migrate}} \le (D-1)\eta v$$

$$t_{\text{scan, days}}(C, D) = \frac{10^6 C}{86400 D \eta v}, \qquad \eta = 0.7,\ v = 300\ \text{MB/s}$$

| Archive | Full scan: two drives | Full scan: three drives |
|---|---|---|
| 1 PB | 27.6 days | 18.4 days |
| 5 PB | 137.8 days | 91.9 days |
| 10 PB | 275.6 days | 183.7 days |
| 20 PB | 551.1 days | 367.4 days |
| 50 PB | 1377.9 days | 918.6 days |

All drives are devoted to scanning in this table. At 10 PB, a yearly complete scan consumes about 10 PB of transfer; with three drives, the one-failure budget then leaves only 3.25 PB/year for all other traffic. At 20 PB, even three healthy drives cannot finish a full scan within one year at the planning rate.

**Consequently, two or three drives can support a very large cold collection only if access and verification demands stay bounded.** They do not make arbitrary archive size operationally equivalent. At 50 PB, a full scan takes roughly 2.5 years even with three drives assigned continuously.

A common library robot, host, power path or catalog can still stop access. The report does not claim a system uptime SLA from drive redundancy. Ejected cartridges do not depend on powered-library availability for retention.

# 5 / Expansion: LTO avoids old-data transformation

The strongest growth advantage is architectural: additional cartridges can accept new video without redistributing the existing archive. Growth of online disk storage may instead require reshaping, repeated disk replacements, additional RAID groups, or changes to pooled storage. These are distinct operations.

| Expansion method | Existing-data work and operational consequence |
|---|---|
| Add a tape cartridge | Write new media and register it in the catalog. Existing cartridges need no rewrite. Additional shelves or library slots may eventually be needed. |
| Linux MD RAID5/6 reshape | Changes stripe geometry; critical regions may require a spare or external backup file to resume safely after a crash. Exact procedure depends on layout and implementation. [7] |
| Expand an OpenZFS RAIDZ group | Reads allocated data and rewrites its layout; preserves redundancy, pauses for failed-disk repair and initiates a scrub afterward. It is not inherently destructive. [8] |
| Add a new RAID group / vdev | Avoids in-place reshape of existing groups. Still adds powered devices and management overhead; a single pooled failure domain may grow. [9,10] |
| Replace all members with larger HDDs | Usually requires several replacement/reconstruction operations before obtaining the larger group capacity; behavior depends on the storage stack. |

## Quantify exposure without inventing an expansion loss rate

$$P(\text{device failure during growth}) = 1 - e^{-\sum_i \lambda_i t_i}$$

$$\text{For equal exposure:} \qquad P = 1 - e^{-m\lambda_H t}$$

Here m drives are involved for t years. With 10 drives, a seven-day operation and assumed $\lambda_H$ = 2%/year, the probability of at least one drive failure during that window is **0.383%**. This is background hardware exposure, not the probability of a failed reshape or data loss, and not necessarily incremental risk caused by the operation.

For an illustrative operation reading and rewriting 1 PB of allocated physical data, traffic is about 2 PB before extra parity, metadata or verification overhead. At an aggregate 1 GB/s this alone occupies about 23.1 days. Real duration depends on the implementation and foreground workload; a multi-group design need not reshape the entire archive.

**Defensible wording:** LTO removes the need to transform old data merely to add capacity. HDD expansion can add prolonged I/O, restart/recovery dependencies and operator risk. A universal percentage for RAID expansion failure is not established by the available evidence.

# 6 / Other HDD drawbacks that matter at scale

## Powered-media energy grows with stored capacity

$$E_H(C) = \frac{8760 N_H(C) P_H}{10^6} \text{ MWh/year}$$

$$E_D = \frac{8760 D P_D}{10^6} \text{ MWh/year}$$

| Retained | HDD idle reference | HDD active reference |
|---|---|---|
| 1 PB | 3.31 MWh/year | 4.10 MWh/year |
| 5 PB | 16.00 MWh/year | 19.82 MWh/year |
| 10 PB | 32.01 MWh/year | 39.63 MWh/year |
| 20 PB | 64.02 MWh/year | 79.26 MWh/year |
| 50 PB | 160.05 MWh/year | 198.15 MWh/year |

IronWolf datasheet reference points are 6.3 W idle and 7.8 W average operating. HPE tape-drive references are 17 W idle and 33 W typical: two drives imply 0.30–0.58 MWh/year; three imply 0.45–0.87. These are operating-point estimates, not hard minimum/maximum bounds. Cartridges themselves draw no electrical power while stored. [2,3]

## Rebuild traffic and maintenance burden

A full 24 TB reconstruction takes 66.7 hours at an effective 100 MB/s target rate, before replacement delays. If eight surviving members are read fully, that reconstruction reads about 192 TB and writes 24 TB. Actual reads vary with implementation and allocated space. At 10 PB, the MTBF model predicts roughly two HDD failures/year; a 2% sensitivity case predicts 11.6. Tape-drive replacement normally does not require reconstructing the archive.

Each full physical HDD scrub contributes 24 TB of reads per drive. Monthly scans add 288 TB/year per drive. Against a specified 550 TB/year workload rating, this leaves about 262 TB/year for all other transfers. Cold-data retention still requires powered hardware and monitoring even when foreground access is small. [1]

## Fault containment and administrative exposure

A cartridge-local failure can be confined to one cartridge if the archive format is independently recoverable. By contrast, losing a top-level data vdev in a ZFS pool loses the pool; independent disk pools can reduce this scope. The correct comparison is architecture-specific, not “one disk failure loses everything.” [10]

Physically offline or write-protected tapes also reduce exposure to online deletion and overwrite. A library-connected RW cartridge is not automatically air-gapped or immutable. Large site events, shared media defects, catalog loss and missing encryption keys can affect many cartridges at once.

Tape has its own lifecycle: controlled storage, inventory, compatibility testing, eventual migration and rare-access latency. Two or three drives are a small service fleet, but a drive that damages tapes could affect multiple mounts; monitor errors and quarantine suspect drives.

# 7 / Proposed architecture and acceptance conditions

## Adopt LTO as the capacity tier, not as the live recording device

Use an online staging tier for active recordings and queries. Consolidate closed video into large sequential archive units, write one LTO copy, and verify it before retiring the staging copy. Maintain two compatible drives as a minimum access configuration, or three where concurrent ingest, retrieval and servicing are important.

The final retained video remains single-copy as requested. Transient staging and replicated catalog/checksum/key metadata do not constitute a second retained video copy. Their resource requirements are outside the device-count and power tables in this report.

| Acceptance item | Required evidence / operating rule |
|---|---|
| Ingest and retrieval | Measure peak and sustained traffic; prove that D−1 drives can meet the workload with the chosen buffering policy. |
| Initial archive integrity | Verify the completed write against source checksums; retain the source until verification succeeds. Test recovery using another compatible drive. |
| Independent recoverability | Keep indexes, manifests and necessary metadata recoverable without depending on unrelated cartridges. Protect catalog and encryption keys separately. |
| Cold-media management | Use manufacturer storage conditions, inventory controls and a documented inspection/verification schedule. [11] |
| Media loss tolerance | Approve the consequences of losing a cartridge or file. The selected q scenario is not an empirical assurance. |
| Migration | Keep compatible readers available and reserve bandwidth for migration to new media. Never overwriting a cartridge does not prohibit migration. |

## Capacity and time must both be budgeted

$$\mathbb{E}[F_H] = \lambda_H \int_0^T N_H(t)\,\mathrm{d}t$$

$$P_{\text{loss, tape}} = 1 - \exp\left[-q \int_0^T N_T(t)\,\mathrm{d}t\right]$$

For the tape expression, $N_T(t)$ is the scheduled cold-media population along the no-failure path. It describes first loss under constant independent hazards. A collection growing to 10 PB over five years has different exposure from 10 PB stored for all five years; final capacity alone is insufficient.

## Decision statement

**LTO is the preferred capacity-growth option for infrequently accessed PB-scale VMS retention when the organization accepts single-copy media risk and can meet its access and verification commitments.** It avoids proportional growth in powered media, minimizes old-data transformation during expansion, and separates drive maintenance from stored-media survival. The recommendation rests on operational and infrastructure advantages, not an unsupported claim of superior single-copy durability.

# 8 / References and calculation notes

Primary sources checked 21 October 2026. Product figures are specifications. Failure counts, energy estimates and throughput tables are calculated from those figures. q, 72-hour HDD restoration, 70% streaming efficiency, 10% allocation reserve and the 2% HDD stress case are assumptions.


[1] Seagate IronWolf Pro product manual, Rev B, July 2025; ST24000NT002, reliability/workload conditions

[2] Seagate IronWolf Pro datasheet; 24 TB model and power reference values

[3] HPE LTO Ultrium Tape Drives QuickSpecs; MTBF, native capacity, speed, error rate and power

[4] HPE LTO-9 Technical Reference Volume 4: Storage (archival)

[5] Schroeder and Gibson, Disk Failures in the Real World, USENIX FAST 2007

[6] LTO Program, LTO-9 UBER Technical Paper, August 2022

[7] mdadm source manual, Grow / RAID-device changes; implementation-specific critical-section recovery

[8] OpenZFS zpool-attach manual; RAIDZ expansion, redundancy, allocated-data rewrite and scrub

[9] OpenZFS zpool-add manual; adding new top-level vdevs

[10] OpenZFS VDEVs documentation; redundancy and pool failure scope

[11] IBM TS4500: Environmental and shipping specifications for LTO cartridges

[12] HPE LTO Ultrium Cartridges QuickSpecs; Q2079A LTO-9 RW media


## Reproduction notes

HDD groups = ceil(C/172.8); HDD count = 10 × groups. Single-copy cartridges = ceil(C/16.2). C is in TB. HDD expected failures/year = count × 0.003504. Tape-drive expected failures/year = D × 0.03504. No hot-spare HDDs are included; adding powered spares adds their exposure and energy.

All probability columns concern the named failure mechanism. An expected device failure is not necessarily an outage; an outage is not necessarily permanent data loss. Expected lost capacity is not the same as the chance of losing any recording. Common-cause failures are not quantified.

RAID6 and RAIDZ2 are both dual-parity designs but have different implementations; the report uses a generic RAID6 failure model and labels OpenZFS-specific operational facts explicitly. The cited MD source branch illustrates documented recovery dependencies, not a universal current default for every array implementation. The HPE archival-life entry was available in indexed documentation; its direct page did not provide extractable body text.